Original Manuscript ID: be7bd75c-daba-4386-ad14-beac2bbcac64

Original Article Title: "AI foundation model choice and false positive variation in Anti Money Laundering transaction screening"

To: Discover Artificial Intelligence Editor

Re: Response to reviewers

Dear Editor,

Thank you for allowing a revision of our manuscript, with an opportunity to address the reviewers' comments.

We are uploading (a) our point-by-point response to the comments (below) (response to reviewers), (b) an updated manuscript with highlighting indicating changes (as "Highlighted PDF"), and (c) a clean updated manuscript without highlights ("Main Manuscript").

Best regards,

Samir Chincholikar
Robin Chawla

Editor, Concern #1

Editor comment:

"Please address the label-dependent aggregation rule by reporting primary results under a label-independent rule and providing sensitivity analyses under alternative tie policies. Also discuss the limitations of case construct validity and statistical independence assumptions."

-Shiping Wang

Author response:

- We agree, and the correction is larger than a change of rule. Decision D4 as preregistered is already label-independent: the decisions log and both preregistration documents specify the modal replicate decision with ties broken toward `flag`, with no reference to the case label. The implemented benign branch required a strict majority, which is an undisclosed deviation. The label-independent rule requested here is therefore the preregistered rule, and we have restored it rather than adopted a new one.
- Under the preregistered rule the false-positive rates are 2.3, 7.7, 37.0, 44.0 and 85.7 percent, a spread of 83.3 pp (95% CI 71.8–93.7) against 82.7 pp as submitted. The spread is 77.7 to 85.3 pp across five aggregation rules and four single-call configurations, with no reversal of model order in any of the nine.
- On construct validity, no independent AML expert adjudication or inter-rater check was performed; labels reflect the generator-defined case templates. On statistical independence, cluster-bootstrap intervals quantify uncertainty across the eight case-template strata, and a cluster-level permutation procedure tests the global null: none of 2,000 permuted statistics reached the observed value (Monte Carlo $p \approx 0.0005$), alongside the corrected case-level Cochran's $Q = 585.0$ ($p \approx 2.7 \times 10^{-125}$).

Author action:

- Section 2.3 was rewritten to state decision D4 as preregistered, to report the deviation in the benign branch, and to make the preregistered label-independent rule primary.
- Section 2.3 now records the departure and its effect on every affected number.
- The Appendix A sentence attributing a label-dependent rule to preregistration was corrected; D4 does not condition on the case label.
- New Table 1 reports false-positive rates, the spread and Cochran's Q under five aggregation rules and four single-call configurations.
- Section 4.2 gained two limitation paragraphs, one on label construct validity and one on the independence assumptions of the primary tests.
- Table 1 retains the deviating rule alongside the preregistered rule so that both sets of numbers remain in the record.

Reviewer #1, Concern #1

Reviewer comment:

The paper evaluates five foundation LLMs for AML screening, finding false-positive rates ranging from 0.3% to 83.0%.

Results are reported accurately with appropriate statistical measures.

Conclusions are well-founded with clear caveats about synthetic data limitations and hypothetical projections.

I recommend that the manuscript be accepted. The study is methodologically sound, results are reported accurately, and conclusions are appropriate.

Author response:

- We thank Reviewer 1 for recommending acceptance.
- The range cited, 0.3 to 83.0 percent, came from a rule that deviated from the preregistration (see Reviewer 3). It is retained in Table 1 as the as-submitted row; the preregistered rule gives 2.3 to 85.7 percent. The finding the reviewer endorses is unchanged.

Author action:

- No change was requested. The correction described under Reviewer 3 updates the figures cited in this report.

Reviewer #2, Cover comments

Reviewer comment:

The paper asks whether the choice of commercial foundation model, rather than the screening task or prompt, materially changes the operating point of an LLM-based AML transaction screener. The authors run five zero-shot foundation models from two providers (three OpenAI, two Google) on an identical, preregistered battery of 600 synthetic transaction cases (300 suspicious across eight FATF-anchored typologies, 300 legitimate "hard negative" cases), crossed with two prompt formulations and two random seeds (12,000 calls total). They report a large spread in false-positive rate across models (0.3%–83.0%), a monotonic within-provider relationship between model tier and false-positive rate, high cross-model disagreement on legitimate cases (85.7%), low chance-corrected agreement on misses, and an illustrative $\sim$197-fold difference in projected daily alert volume under a low-prevalence scenario. The paper frames these findings as a model-risk/governance issue rather than a model-capability issue, and supplies a full reproducibility package.

This is a carefully designed and quite transparent empirical study. The core idea — treating foundation-model identity as the independent variable while holding the task and cases fixed, and reframing the result in model-risk-governance terms — is a useful contribution to a literature that often focuses on identifying the best model. The statistical treatment is appropriate for the paired design, and the reproducibility package is appreciated.

I recommend minor revisions, focused mainly on (a) sharpening the originality claim (b) clarifying a few methodological choices that affect the headline numbers, and (c) some robustness check.

Please find the comments do be addressed in the uploaded file.

Author response:

- We thank Reviewer 2 for the detailed report. The thirteen numbered comments in the attachment are answered individually below, in the order the attachment gives them.
- One correction to the call count, which is ours. The design specified 12,000 calls and the frozen corpus holds 12,120: 111 GPT-4o-mini cases carry a fifth retried call and one GPT-4.1-mini case carries three after an interface error. The manuscript disclosed the retries qualitatively but never gave the realized total. The 111 retries were made because the GPT-4o-mini terse-prompt seed-0 pass was re-issued after OpenAI rate limiting interrupted it, so cases that had already returned were called again; the remaining 9 calls are ERROR rows recording the rate-limit failures, which are excluded from the vote. Restricting the

vote to the four designed calls changes one GPT-4o-mini decision (85.7% to 86.0%; spread 83.3 to 83.7 pp), reported as a sensitivity analysis in Section 2.2.

Author action:

- Section 2.2 now reports the designed count of 12,000 and the realized count of 12,120, with the retry and error breakdown. Data availability now gives the realized count.

Reviewer #2, Concern #1

Reviewer comment:

The central claim is credible but should be stated more precisely. The authors state they are "the first study to control foundation-model identity as the experimental variable in AML transaction screening while fixing the task and transaction cases". Within AML specifically this is plausible and, to my knowledge, not previously done in this controlled way. However, the general phenomenon being measured — that different LLMs disagree substantially on identical inputs, and that prompt-induced variance is smaller than model-induced variance — has precedent in the broader LLM-evaluation literature. I would ask the authors to sharpen the novelty claim to "first domain-specific, governance-framed controlled measurement of this effect in AML screening" rather than implying the phenomenon itself is new.

Author response:

- We agree and adopt the reviewer's formulation. Cross-model disagreement on identical inputs is established in the model-evaluation literature and we should not have implied otherwise.
- The contribution we claim is the domain-specific, governance-framed controlled measurement in AML screening, together with the translation into alert burden.

Author action:

- The contribution sentence in Section 1 now claims the first preregistered, governance-framed controlled measurement of this effect in AML screening (the reviewer's formulation, narrowed to the preregistered design).
- Contribution 2 in Section 1 now cites the LLM-evaluation work that established the general phenomenon [26, 39].
- The same narrowing was applied to the abstract and to the first paragraph of the Conclusion.

Reviewer #2, Concern #2

Reviewer comment:

The governance reframing is the paper's strongest and most original contribution. Treating model-tier substitution and provider-managed '-latest' aliases as first-class model-risk events, and proposing a standing "operating-point probe" as a monitoring control, is a useful and new operationalisation for this domain. This contribution stands on its own even if the underlying "models disagree a lot" finding were less novel, and I'd encourage the authors to foreground it even more.

Author response:

- We thank the reviewer and agree the reframing should carry more weight in the paper's structure than it

did.

- Foregrounding it also addresses Concern #1, since the governance operationalisation does not depend on the disagreement phenomenon being new.

Author action:

- The operating-point probe was promoted from a passage in the Discussion to its own numbered subsection, stating the control, its trigger conditions, and what is monitored.
- Model-tier substitution and provider-managed alias drift are now named as model-risk events in the abstract and the Conclusion.
- The governance contribution is now listed first among the contributions in Section 1.

Reviewer #2, Concern #3

Reviewer comment:

A natural extension that bears on originality. The study covers only OpenAI and Google models, while the Generative AI declaration notes that Anthropic Claude was used for coding assistance. Given the paper's thesis is about provider dependence and model-swap risk, omitting other providers leaves the "provider effect vs. tier effect" distinction slightly underdetermined, a sentence explaining the provider selection (cost, access, or scope reasons) would help.

Author response:

- The reviewer is right that the selection was never explained, and our records show it was not a two-provider design. A third family, xAI Grok-4.5, was preregistered in the confirmatory grid. Two logged decisions reduced it: D17 dropped `gemini-pro` for thinking-mode incompatibility and cost, and D19 dropped `xai-grok` because the account was out of credits. The five-model grid is documented attrition. Because tier is defined only within providers, a provider main effect cannot be separated from the tier effect.
- Anthropic Claude appears in the generative-AI declaration solely as a coding assistant and was never an evaluated system. The tier effect remains identified within each provider family separately.
- A related correction belongs here. The submitted Methods stated that the model served behind each Google alias is recorded in the archived per-call metadata. It was not: the capture rows, probe receipts and freeze receipts record only the alias strings. The Limitations state the position correctly and the Methods sentence overstated it.

Author action:

- Section 2.2 now reports the preregistered three-family grid and the two decisions that reduced it (D17, D19).
- A sentence clarifies that Anthropic Claude was a coding assistant and not an evaluated model.
- The Methods sentence claiming that the resolved Google model version is recorded in the archived metadata was removed and replaced with a statement that only the alias was captured.
- The Limitations state that the provider main effect cannot be separated from the tier effect, and name a third-provider replication as the natural extension.
- Data availability now states that, for the Google models, the served-model field records the alias only.

Reviewer #2, Concern #4

Reviewer comment:

Construct validity of the case battery. The false-positive/miss-rate estimates rest entirely on the correctness of the synthetic labels. The Methods section describes how cases were generated but does not state whether the "legitimate hard negative" and "suspicious" labels were reviewed or validated by an independent AML domain expert. I'd ask the authors to clarify whether any expert adjudication or inter-rater check was performed on a sample of the battery, and if not, to say so explicitly as a limitation.

Author response:

- No expert adjudication and no inter-rater check were performed on any part of the battery. Labels are assigned by the generator at construction time: a case is suspicious because it was built from one of the eight FATF-anchored typology templates, and legitimate because it was built from a hard-negative template.
- Our rates therefore measure agreement with the generator's construction, not with an expert's assessment of real transactions. A model penalised for flagging a hard negative may be responding to a feature an experienced analyst would also find suspicious.
- We did not commission a sample adjudication during revision, because an adjudication by the authors would not be independent. We prefer to state the gap than to report a weak substitute for it.

Author action:

- Section 2.1 now states that labels are true by construction and that no independent expert review or inter-rater reliability check was conducted.
- Section 4.2 gained a limitation paragraph explaining that the rates are measured against template-defined rather than expert-adjudicated ground truth.
- An expert-adjudicated subsample is named as required future work.

Reviewer #2, Concern #5

Reviewer comment:

The tie-breaking rule in the aggregation procedure (Appendix A, decision D4) deserves more scrutiny. The rule resolves 2–2 replicate splits "with reference to the known case label" (a tie counts as not a false positive for benign cases, and counted as caught for suspicious cases). This rule affects 132/1500 (8.8%) of benign and 48/1500 (3.2%) of suspicious model-case decisions, which is not negligible relative to some of the reported rates. I would ask for:

a breakdown of how many ties occurred per model, since if ties cluster in particular models, the label-referenced tie-break could differentially inflate or deflate that model's reported rate;

a sensitivity analysis reporting the headline spread (82.7 pp) and the cross-model Q-test under an alternative, label-agnostic tie rule, to show the qualitative conclusion is not an artifact of D4.

Author response:

- Ties do cluster. On legitimate cases the counts are 6, 20, 40, 58 and 8 for Gemini Flash, GPT-4o, GPT-4.1-mini, Gemini Flash-Lite and GPT-4o-mini, and on suspicious cases 19, 15, 10, 4 and 0. They concentrate where a model sits near its own decision boundary, so the label-agnostic rule raises Gemini

Flash-Lite by 19.3 pp and GPT-4.1-mini by 13.3 pp but Gemini Flash by only 2.0 pp. The deviation compressed the middle of the distribution, not the extremes.

- The spread is therefore nearly unaffected: 83.3 pp against 82.7 pp, with Cochran's $Q = 585.0$ on 4 df ($p = 2.7 \times 10^{-125}$) against a corrected $Q = 657.8$ as submitted (see Concern #6). One pairwise comparison strengthens: Gemini Flash versus GPT-4o was not significant as submitted after correction and is now significant at adjusted $p = 4.0 \times 10^{-3}$.
- Checking this comment led to a more serious finding, which we report in full. The label-dependent rule is not what decision D4 preregisters: D4 and both preregistration documents specify ties broken toward `flag` without reference to the case label, so the benign branch as implemented was an undisclosed deviation. The label-agnostic figures the reviewer asks us to test are the preregistered figures, and the deviation understated every published false-positive rate.

Author action:

- Section 2.3 and Table 5 report per-model tie counts on both labels; Table 1 gives the resulting per-model change in false-positive rate.
- The sensitivity table reports the spread and Cochran's Q under five aggregation rules and four single-call configurations.
- Section 2.3 states decision D4 as written and records the departure from it.
- The pairwise McNemar table is now reported under the preregistered rule.

Reviewer #2, Concern #6

Reviewer comment:

Statistical independence assumptions. McNemar's test and Cochran's Q assume the 300 legitimate cases are exchangeable/independent units. Since cases are synthetically generated from a shared typology structure, some correlation across cases seems plausible. The authors use cluster bootstrapping over the eight typology strata for the secondary uncertainty estimates but not, as far as I can tell, for the primary McNemar/Cochran tests. Given how strong the effect is ($P = 2 \times 10^{-27}$), this is unlikely to change the qualitative conclusion, but a brief comment on why case-level inference is appropriate for the primary test would be useful.

Author response:

- The reviewer has read the code correctly: the cluster bootstrap was applied to the secondary interval estimates only. Rather than defend case-level inference we extended the cluster-robust treatment to the primary test, where a permutation that applies a single permutation of model labels to all cases within each of the eight case-template strata gives Monte Carlo $p \approx 0.0005$ (plus-one correction) under either rule.
- Adjacent-tier contrasts under the restored rule are 29.3 pp (95% CI 12.6–51.2), 48.7 pp (25.8–72.5) and 41.7 pp (19.7–65.2). All exclude zero, and all are materially wider than case-level inference would give, so we report the wider ones.
- On the p-value the reviewer quotes, the revision audit found two errors in our statistics. First, `stats_tests.json` recorded it as 0.0 because the code computed one minus the chi-squared CDF, which underflows. Second, the Q statistic itself was understated fivefold, because the code omitted the factor k from the numerator. Under the preregistered rule the corrected statistic is $Q = 585.0$, $p = 2.7 \times 10^{-125}$; the submitted $Q = 131.6$ should have been 657.8. The permutation result is unchanged, because a constant factor does not alter permutation ranks. Both errors understated the evidence; neither changes a conclusion.

Author action:

- Section 2.4 now states the independence assumption, notes that cases sharing a case template are not independent, and reports the cluster-level permutation test as the primary inference.
- Adjacent-tier contrasts are reported with cluster-robust bootstrap intervals over the eight strata.
- The analysis code was corrected to use the chi-squared survival function. The capsule was re-run and re-hashed.
- The Cochran's Q implementation was corrected, all Q values were recomputed from the frozen decisions and cross-checked against an independent implementation, and Section 2.4 records the correction.

Reviewer #2, Concern #7

Reviewer comment:

"Model tier" is treated ordinally but tested only descriptively. The paper observes a monotonic FPR decrease with tier within each provider and calls this a "consistent pattern,". I'd suggest adding a formal trend test to support the word "monotonic," or otherwise soften "monotonic" to "monotonically decreasing in the two families studied" (which is closer to what is actually shown with only 2–3 tiers per provider).

Author response:

- We take both halves of the suggestion: we add a formal test and soften the wording. We use adjacent-tier paired McNemar tests rather than a Cochran–Armitage trend test, because a trend test would treat the same 300 cases as independent groups and overstate the evidence.
- All three adjacent-tier comparisons run in the predicted direction and remain significant after correction, with discordant counts 2 and 90, 3 and 149, and 0 and 125.
- The ordering also survives the change of rule, which we could not assert before: it holds under the submitted rule, under the preregistered rule, and in all four single-call configurations.

Author action:

- "Monotonic" is now qualified as holding within each of the two provider families studied, in the abstract, Section 3.2 and the Conclusion.
- Adjacent-tier paired McNemar results with cluster-robust interval estimates are reported in Section 3.2, with a sentence explaining why a paired test is preferred to Cochran–Armitage in this design.

Reviewer #2, Concern #8

Reviewer comment:

Operational projection is a single point, not a curve. The 197-fold alert-volume claim rests on one prevalence assumption (0.1%) and one volume assumption (1,000,000/day). The paper is appropriately cautious that this is illustrative, but since alert volume and precision both scale nonlinearly with prevalence, a supplementary figure sweeping prevalence over a plausible range (e.g., 0.01%–1%) would let readers see whether the relative ordering of models (and the $\sim$200-fold spread) is stable across the plausible range, or whether it is sensitive to the specific 0.1% choice.

Author response:

- We swept prevalence from 0.01 to 1.0 percent. The ordering of the five models by alert volume is identical at every prevalence.
- The magnitude is sensitive. Under the preregistered rule, the highest-to-lowest ratio is 36.6 at 0.01 percent, 35.4 at 0.1 percent and 26.9 at 1.0 percent, because true positives take a growing share of the queue for low-false-positive models as prevalence rises.
- The magnitude also depends on the aggregation rule. Under the submitted rule the ratio was 197 at 0.1 percent; it falls to 35 because Gemini Flash's false-positive rate rises from 0.3 to 2.3 percent under the preregistered rule. The four deployable single-call configurations give 36- to 196-fold.
- Precision ranges from 0.4 to 27.6 percent for Gemini Flash and from 0.01 to 1.2 percent for GPT-4o-mini.

Author action:

- New Fig. 4 sweeps prevalence from 0.01 to 1.0 percent on a logarithmic axis, with alert volume per model and a companion precision panel.
- Section 3.3 scopes the ratio to its prevalence and configuration, gives the swept range (27- to 37-fold), and states that the model ordering is invariant.
- The abstract reports 36- to 196-fold across deployable configurations and about 35-fold under the preregistered aggregate.

Reviewer #2, Concern #9

Reviewer comment:

Zero-shot-only scope. This is a deliberate and well-justified design choice (isolating the base-model contribution before task-specific tuning), and the limitation is honestly stated. No change needed, but I'd suggest one added sentence in the Discussion speculating on whether calibration/threshold-tuning would be expected to compress the cross-model spread (since institutions could in principle recalibrate each model to a common operating point) or whether the paper's evidence (e.g., the persistence of ordering across two prompts) suggests the divergence is more structural than a simple decision-threshold shift. This matters for how alarming the governance implication should be considered.

Author response:

- The elicitation returns a binary label rather than a score, so there is no decision threshold to tune without changing the elicitation itself. Recalibrating these screeners to a common operating point is not a post-processing step.
- The ordering persists across both prompt formulations and all four single-call configurations, with the spread between 80.3 and 85.3 pp in every one, which is more stability than a pure threshold shift would predict.
- This is not a demonstration that score-level calibration could not compress the spread. A scored elicitation with per-model thresholds is the experiment that would settle it, and we state the implication conditionally.

Author action:

- The Discussion gained a passage noting that the binary elicitation leaves no threshold to tune, that the persistence of ordering is consistent with structural rather than threshold divergence, and that a scored elicitation is required to settle the question.
- The governance implication is now stated conditionally.

Reviewer #2, Minor Point #10

Reviewer comment:

Please define "model tier" operationally in the Methods (currently it is defined via "provider's contemporaneous capability/product and price positioning," which is reasonable but could usefully cite the specific list-price/capability source alongside the models.yaml registry reference).

Author response:

- We agree, and verifying the source changed what we can defensibly write. The three OpenAI prices in the archived registry are the published list prices, confirmed against the OpenAI pricing documentation: $2.50 and $10.00 per million input and output tokens for GPT-4o, $0.40 and $1.60 for GPT-4.1-mini, and $0.15 and $0.60 for GPT-4o-mini.
- The two Google figures are not list prices. The registry states that they were set conservatively so the spend cap would bind early and are overwritten with measured cost, and the July 2026 list prices are not recoverable because the models were reached through `-latest` aliases whose resolved versions were not captured.
- We will not cite a price we cannot substantiate. Tier is defined by published price positioning, cited for OpenAI, and by the measured per-call cost in the archived ledger for all five models, which reproduces the tier ordering from the deposit without relying on an unverifiable figure.

Author action:

- Section 2.4 now gives the operational definition of tier, the verified OpenAI list prices with their source and access date, and the measured per-call costs from the archived ledger for all five models.
- The registry reference now states that its Google price fields are conservative budgeting placeholders and not list prices.
- The text repeats that tier is used only within provider families and not as a cross-provider capability scale.

Reviewer #2, Minor Point #11

Reviewer comment:

The abstract states "an 82.7 percentage-point spread" while the Results also report a 95% CI of 69.2–95.3 pp for that spread; consider including the CI in the abstract as well, since a >25-point CI width is itself informative.

Author response:

- We agree that the width is itself a finding.
- Under the restored rule the abstract reports 83.3 pp with a 95% confidence interval of 71.8 to 93.7, with a clause stating that the interval is a cluster bootstrap over eight case-template strata.

Author action:

- The abstract now reports the spread with its cluster-bootstrap 95% confidence interval.
- A clause states that the interval is a cluster bootstrap over eight case-template strata, so the width is not read as sampling error over 300 independent cases.

Reviewer #2, Minor Point #12

Reviewer comment:

Consider explicitly stating in the Limitations that only two providers/five models were tested, so the paper's own headline "foundation-model choice" language should be read as scoped to the evaluated model set, not as a universal claim about all foundation models (the paper mostly does this correctly, but the Conclusion's phrasing is slightly more general than the evidence supports).

Author response:

- We agree, including on the Conclusion being the place where the phrasing outruns the evidence.
- Five models from two providers support a claim about the evaluated set and about the existence of the effect, not about foundation models in general.

Author action:

- The Limitations state that the evidence covers five models from two providers and that the headline language is scoped to the evaluated set.
- The Conclusion was reworded to replace the general phrasing with the scoped claim, and names the generalisation as a hypothesis for broader replication.

Reviewer #2, Minor Point #13

Reviewer comment:

There are quite many typos such as numbers inserted across brackets in many paragraphs. Formatting also could be improved.

Author response:

- We identified two causes: misplaced citations (below) and lost en dashes (last action). In three places a citation sat immediately before the closing em dash of a parenthetical aside, so the reference number renders inside the aside and interrupts the phrase, as in "a known source of variation in LLM behaviour [39]—each case was tested". The three occurrences are in Sections 1, 2.2 and 2.4.
- The defect is masked in the compiled PDF but visible in the Word rendering, which is why we missed it. We now proofread the Word rendering as well as the PDF, and an automated check rejects any citation falling inside an em-dash aside.

Author action:

- The three citations were moved outside their em-dash asides, so each reference number now precedes the aside rather than interrupting it.
- The design equation was rewritten as prose, which renders correctly in both the PDF and the Word file.
- An automated check now rejects any citation falling between two em dashes in the converted Word file; it was tested by reintroducing one defect and confirming it is caught.
- Numeric formatting, table alignment and thousands separators were made consistent throughout.
- En dashes in numeric ranges were lost in the PDF text layer, so "[1.1–4.7]" read as "[1.14.7]". The PDF now embeds Unicode mappings for dashes and ligatures.

Reviewer #2, Recommendation

Reviewer comment:

Minor revision. The core empirical contribution — a controlled, paired, preregistered demonstration that foundation-model identity alone produces large, statistically robust shifts in AML screening operating points, reframed as a model-risk governance issue — is original and well-executed. The requested revisions are addressable without new data collection, since the reproducibility package already contains everything needed to rerun the relevant analyses under alternative assumptions.

Author response:

- We confirm the reviewer's expectation. Every revision was carried out by re-analysing the frozen corpus, with no new data collection and no new interface calls.
- The re-analysis re-derives the operating points from the frozen call records without importing the modules used for the original submission, so the recomputed figures are an independent check rather than a re-run. One error survived that re-derivation: Cochran's Q was understated fivefold in both the submitted and first revised analyses (Concern #6). It is now corrected and cross-checked against an independent implementation.

Author action:

- The reproducibility capsule was extended with the aggregation-sensitivity analysis and re-hashed, so the alternative tie policies are reproducible from the archive.

Reviewer #3, Overall assessment

Reviewer comment:

This manuscript investigates whether foundation-model choice materially changes the operating point of LLM-based anti-money-laundering transaction screening when the task and transaction cases are held constant. The topic is timely and the controlled cross-model design, paired statistical testing, reproducibility framework, and translation of false-positive rates into operational alert burden are valuable strengths. However, several methodological issues should be addressed before the conclusions can be fully supported, particularly the per-case aggregation procedure used to combine replicate model decisions.

Author response:

- We thank Reviewer 3 for identifying the aggregation procedure as the issue on which the conclusions depend. We agree and have made it the central revision.
- The report states that several methodological issues should be addressed but sets out one major comment, which we answer in full below. We would welcome the opportunity to address any further comments intended to accompany this report.

Author action:

- The actions are those listed against the major comment below.

Reviewer #3, Concern #1

Reviewer comment:

Major comment — label-dependent aggregation rule:

The preregistered aggregation rule resolves 2–2 replicate splits differently depending on the known ground-truth label: a tie on a legitimate case is treated as legitimate, whereas a tie on a suspicious case is treated as detected. Although this rule was preregistered and transparently disclosed, it uses information that would not be available in deployment and therefore may produce an optimistically biased evaluation operating point. This is especially important because ties affected a non-trivial number of model–case decisions. I recommend reporting the primary results under a label-independent aggregation rule and providing sensitivity analyses under alternative tie policies, such as treating ties as abstentions, always escalating ties, or analysing each prompt configuration separately. The authors should also clarify whether the headline false-positive and miss rates represent deployable model configurations or an evaluation-only aggregation procedure.

Author response:

- We accept the criticism. The rule used the ground-truth label to choose the tie direction, both directions favoured the model, and that is an evaluation artifact rather than a property of any deployable screener. It is also not the preregistered rule: decision D4 and both preregistration documents specify ties broken toward `flag` with no reference to the case label, so the benign branch as implemented was an undisclosed deviation, and we report it as such. We have restored the preregistered rule as primary, which is also the conservative choice on the false-positive axis and corresponds to routing a split panel to a human reviewer.
- We ran each of the three sensitivity analyses named. Ties as abstentions give a spread of 84.9 pp on each model's retained cases, with Cochran's Q computed on the 189 cases all five models decided, always escalating gives 83.3 pp, each prompt configuration separately gives 85.3, 85.0, 80.3 and 81.3 pp, and unanimity gives 77.7 pp. No pair of models reverses order.
- On the final question: the four-call aggregate is an evaluation-only construct, since no institution would issue four calls per transaction and take a majority. The deployable quantity is a single call, where the false-positive rate runs from 0.3 to 2.3 percent for Gemini Flash and from 80.7 to 87.7 percent for GPT-4o-mini. The deviation understated every false-positive rate and left miss rates unchanged. A label-independent strict-majority rule reproduces the submitted false-positive rates but raises miss rates from 12.0, 3.3, 3.7, 0.7 and 0.0 percent to 18.3, 8.3, 7.0, 2.0 and 0.0 percent.

Author action:

- Section 2.3 states decision D4 as preregistered and makes it primary.
- Section 2.3 records the departure in the benign branch and every number it affected.
- The Appendix A sentence attributing a label-dependent rule to preregistration was corrected.
- New Table 1 reports false-positive rates, the spread and Cochran's Q under five aggregation rules and four single-call configurations.
- Section 2.3 and Appendix A state that the aggregate is an evaluation-only construct; Table 1 reports the single-call configurations as the deployable operating points.
- Table 1 retains the deviating rule alongside the preregistered rule.